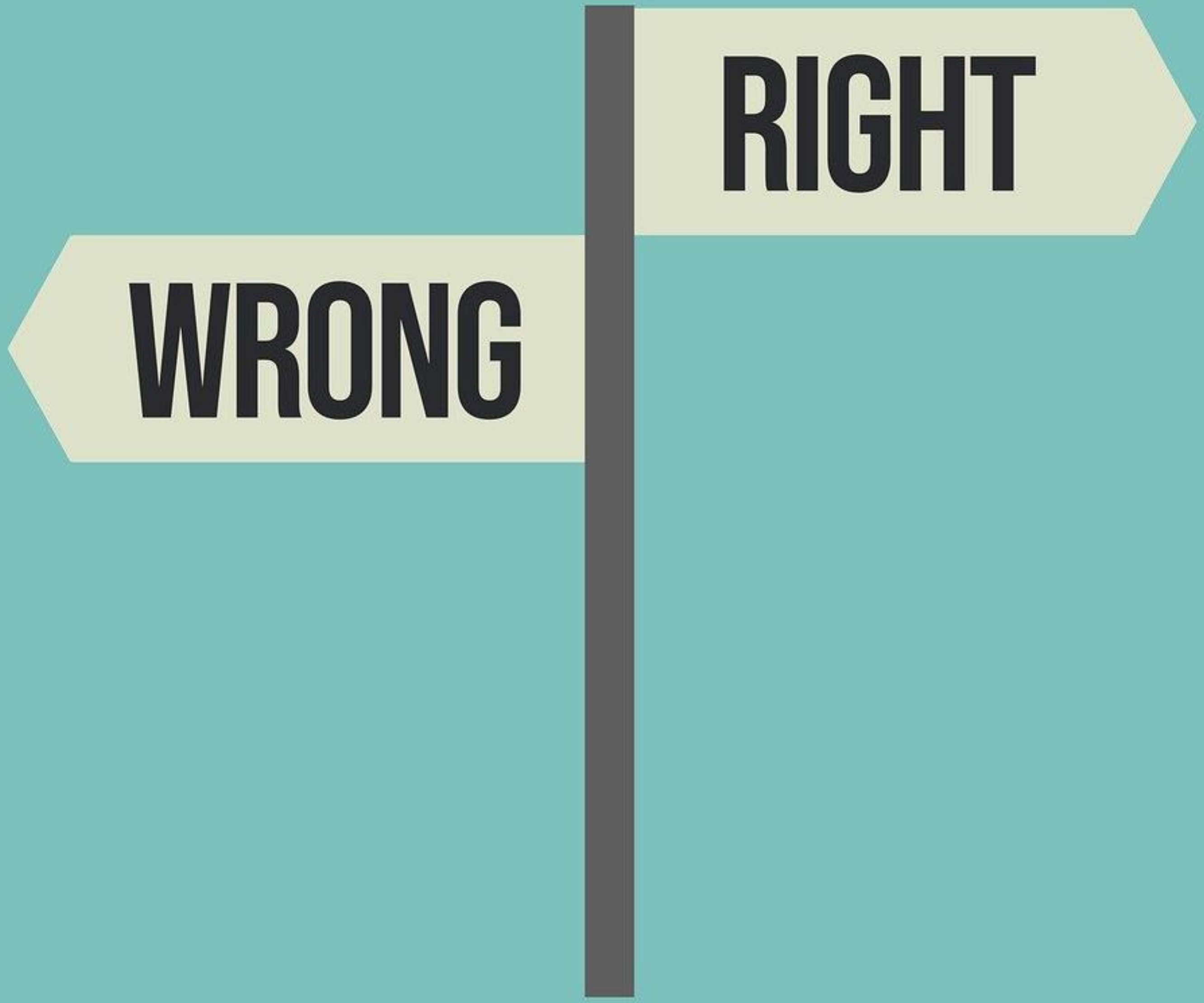
Ethical and Legal Considerations

Final Year Project: 6COSC023W & 6CSEF002W

2025/26

Frantzeska Kolyda MPhil SFHEA FBCS


RIGHT
WRONG

Session Part I

Ethics in Computing

"Ethics permeates the design of almost every computer system or algorithm that's going out in the world.

Prof Barbara Grosz – Harvard University

- NOT something that you think about after you've done your 'real' computer science work.

If you only take away one thing from this session:

To recognise responsibilities that go beyond the implementation of IT systems

- We need to consider not merely what systems we ***could*** build, but whether we ***should*** build those systems and how we should design those systems.

Imagine every time a computer scientist logs on to write an algorithm or build a system, a message will flash across the screen that asks:

> "*Have you thought about the ethical implications of what you're doing?*

Prof Barbara Grosz – Harvard University

“Ethical reasoning is an essential skill for today’s computer scientists.

- Understanding human biases first is important in understanding algorithmic bias.

- Understanding unconscious bias

Autonomous systems

- Make informed decisions for themselves in complex environments.

- The systems can have a physical manifestation such as a vehicle or a robot or be purely software based, such as a financial trading algorithm or a medical decision-support system.

Autonomous systems

- There are a range of ethical considerations throughout the phases of inception, design and deployment of autonomous systems.

Autonomous systems

- At the **inception** stage, the benefits from the autonomous system must be clearly articulated.
- In the **design** phase, decisions will need to be made collaboratively to resolve moral ambiguity and inform the degree to which the system is transparent, unbiased and respects privacy.
- When it comes to **deployment**, the public will need to be consulted and the right governance will need to be put in place to make a judgement about whether or not the benefits should be realised despite uncertainty about the risk of harm or asymmetries in benefits.

Volkswagen America's CEO blames software engineers for emissions cheating scandal

/ Fixing the
least 1-2 yea

Volkswagen emissions scandal

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Volkswagen to pay out £193m in 'dieselgate' settlement

25 May 2022

Diesel emissions scandal

GETTY IMAGES

Volkswagen is to pay £193m to more than 90,000 drivers in England and Wales after it settled a High Court claim over the installation of emissions cheating devices in its vehicles.

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Carmakers go on trial over emissions cheat claims

GETTY IMAGES

There was a small protest by the group Mums for Lungs and outside the High Court as the trial opened

Emer Moreau, Business reporter and **Theo Leggett**, Business correspondent

13 October 2025

Updated 13 October 2025

A major lawsuit against five leading carmakers accused of cheating on emissions tests has opened at the High Court in London.

The trial is the latest chapter of what has become known as the "dieselgate" scandal, with the companies facing allegations they used illegal software to allow their cars to reduce emissions of harmful gases under test conditions.

The court heard on Monday that car manufacturers decided they would "rather cheat than comply with the law" over vehicle emissions.

The five carmakers - Mercedes, Ford, Peugeot/Citroën, Renault and Nissan - all deny the accusations.

video


THE CONVERSATION
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BST

Facebook's psychological study ends up testing users' trust

Experimental evidence of massive-scale emotional contagion through social networks

Adam D. I. Kramer, Jamie E. Guillory, and Jeffrey T. Hancock

PNAS June 17, 2014 111 (24) 8788-8790; published ahead of print June 2, 2014 https://doi.org/10.1073/pnas.1320040111

Edited by Susan T. Fiske, Princeton University, Princeton, NJ, and approved March 25, 2014 (received for review October 23, 2013)

View Full Text

This article has corrections. Please see:

Editorial Expression of Concern: Experimental evidence of massivescale emotional contagion through social networks

Correction for Kramer et al., Experimental evidence of massive-scale emotional contagion through social networks

SEF002W

Editorial Expression of Concern and Correction

PSYCHOLOGICAL AND COGNITIVE SCIENCES

PNAS is publishing an Editorial Expression of Concern regarding the following article: "Experimental evidence of massive-scale emotional contagion through social networks," by Adam D. I. Kramer, Jamie E. Guillory, and Jeffrey T. Hancock, which appeared in issue 24, June 17, 2014, of *Proc Natl Acad Sci USA* (111:8788–8790; first published June 2, 2014; 10.1073/pnas.1320040111). This paper represents an important and emerging area of social science research that needs to be approached with sensitivity and with vigilance regarding personal privacy issues.

Questions have been raised about the principles of informed consent and opportunity to opt out in connection with the research in this paper. The authors noted in their paper, "[The work] was consistent with Facebook's Data Use Policy, to which all users agree prior to creating an account on Facebook, constituting informed consent for this research." When the authors prepared their paper for publication in PNAS, they stated that: "Because this experiment was conducted by Facebook, Inc. for internal purposes, the Cornell University IRB [Institutional Review Board] determined that the project did not fall under Cornell's Human Research Protection Program." This statement has since been confirmed by Cornell University.

Obtaining informed consent and allowing participants to opt out are best practices in most instances under the US Department of Health and Human Services Policy for the Protection of Human Research Subjects (the "Common Rule"). Adherence to the Common Rule is PNAS policy, but as a private company Facebook was under no obligation to conform to the provisions of the Common Rule when it collected the data used by the authors, and the Common Rule does not preclude their use of the data. Based on the information provided by the authors, PNAS editors deemed it appropriate to publish the paper. It is nevertheless a matter of concern that the collection of the data by Facebook may have involved practices that were not fully consistent with the principles of obtaining informed consent and allowing participants to opt out.

Inder M. Verma
Editor-in-Chief

www.pnas.org/cgi/doi/10.1073/pnas.1412469111

PSYCHOLOGICAL AND COGNITIVE SCIENCES

Correction for "Experimental evidence of massive-scale emotional contagion through social networks," by Adam D. I. Kramer, Jamie E. Guillory, and Jeffrey T. Hancock, which appeared in issue 24, June 17, 2014, of *Proc Natl Acad Sci USA* (111:8788–8790; first published June 2, 2014; 10.1073/pnas.1320040111).

The authors note that, "At the time of the study, the middle author, Jamie E. Guillory, was a graduate student at Cornell University under the tutelage of senior author Jeffrey T. Hancock, also of Cornell University (Guillory is now a postdoctoral fellow at Center for Tobacco Control Research and Education, University of California, San Francisco, CA 94143)." The author and affiliation lines have been updated to reflect the above changes and a present address footnote has been added. The online version has been corrected.

The corrected author and affiliation lines appear below.

Adam D. I. Kramer[a,1], Jamie E. Guillory[b,2], and Jeffrey T. Hancock[b,c]

[a]Core Data Science Team, Facebook, Inc., Menlo Park, CA 94025; and Departments of [b]Communication and [c]Information Science, Cornell University, Ithaca, NY 14853

[1]To whom correspondence should be addressed. Email: akramer@fb.com.

[2]Present address: Center for Tobacco Control Research and Education, University of California, San Francisco, CA 94143.

www.pnas.org/cgi/doi/10.1073/pnas.1412583111

CORRECTION

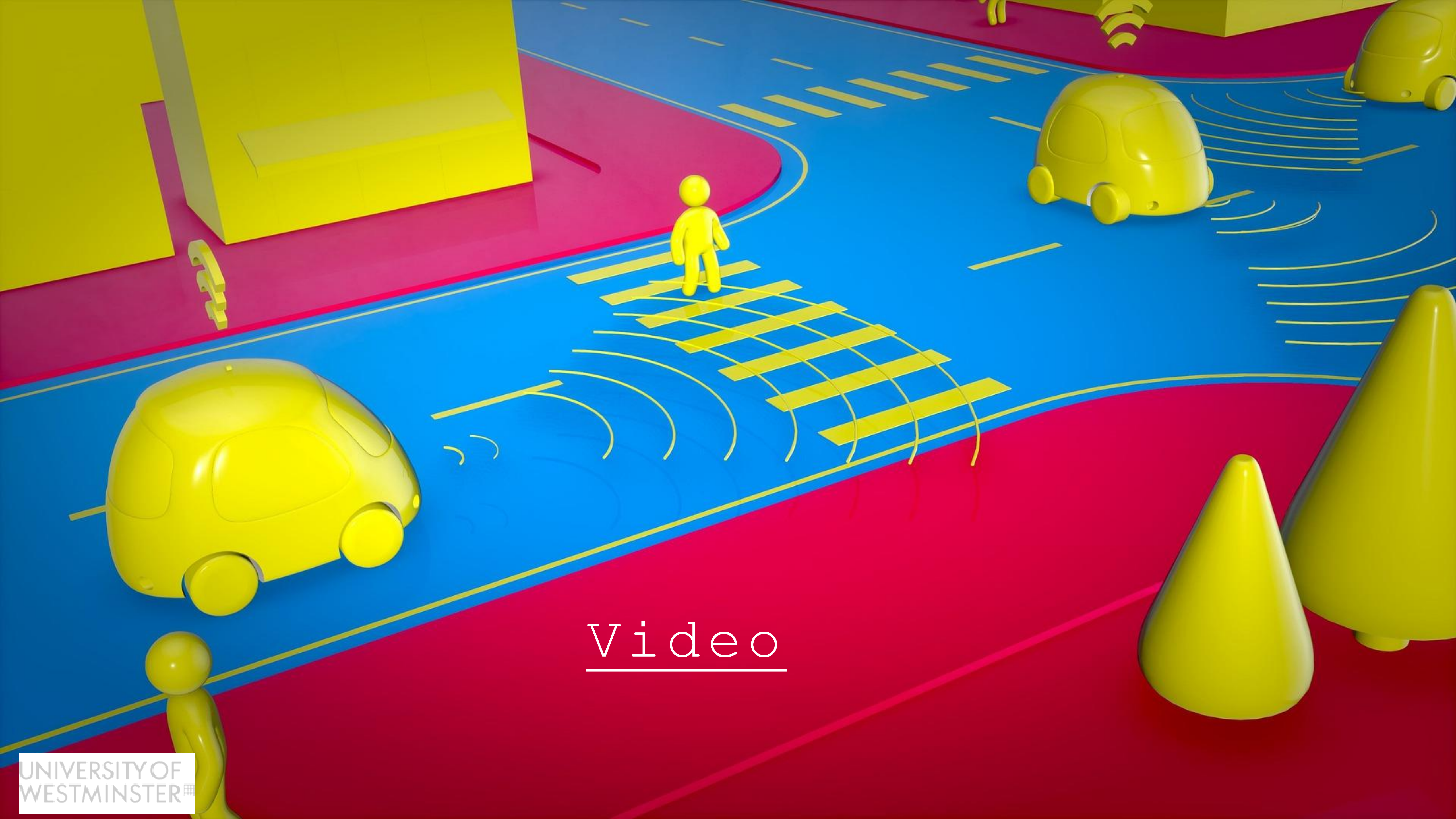
Facebook says it 'unintentionally uploaded' 1.5 million people's email contacts without their consent.

- *A security researcher recently noticed Facebook was asking some new users to provide their email passwords when they signed up — a move widely condemned by security experts.*
- *It was then discovered that if you entered your email password, a message popped up saying it was "importing" your contacts without asking for permission first.*
- *Facebook has now revealed that it "unintentionally" grabbed 1.5 million users' data, and is now deleting it (18th April 2019).*
- Read the story

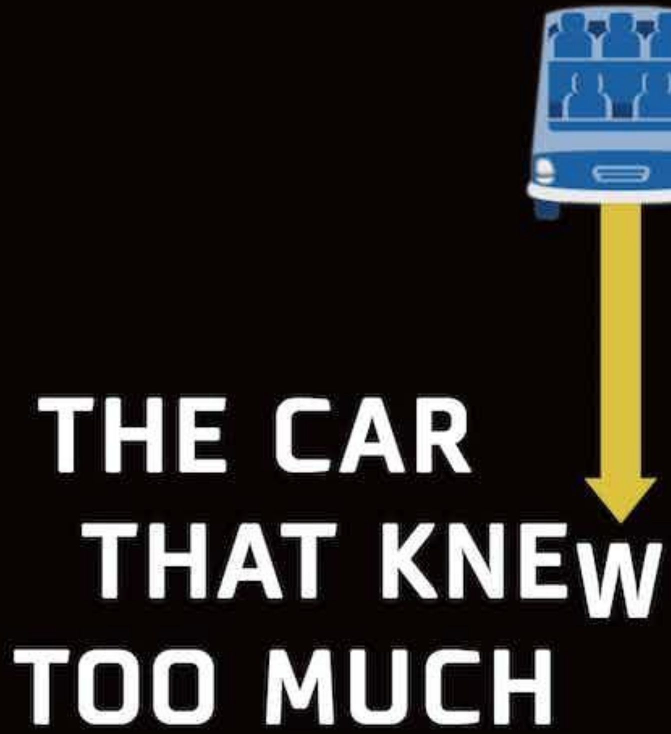
When there is resistance and scrutiny, there can be change

- Facebook (now Meta) deleted a billion faceprints around the time of a [US] $650 million settlement after they faced allegations of collecting face data to train AI models without the expressed consent of users.

video



Video



> *Boosters for driverless cars argue that they will be in fewer accidents than human-driven cars. It's up to humans to decide how many fatal accidents we will allow these cars to have.*

Amazon scraps secret AI recruiting tool that showed bias against women

Video

- *"SAN FRANCISCO (Reuters) - Amazon.com Inc's machine-learning specialists uncovered a big problem: their new recruiting engine did not like women.*
- *The team had been building computer programs since 2014 to review job applicants' resumes with the aim of mechanizing the search for top talent, five people familiar with the effort told Reuters. Automation has been key to Amazon's e-commerce dominance, be it inside warehouses or driving pricing decisions. The company's experimental hiring tool used artificial intelligence to give job candidates scores ranging from one to five stars - much like shoppers rate products on Amazon, some of the people said".*
- [Jeffrey Dastin, Reuters, October 2018]


Video
UNIVERSITY OF
WESTMINSTER


NETFLIX
CODED BIAS
Coded Bias
2020 | 12 | 1h 25m | Science & Nature Documentaries
This documentary investigates the bias in algorithms after M.I.T. Media Lab researcher Joy Buolamwini uncovered flaws in facial recognition technology.

Harm or negative consequences caused by ML systems

- Barocas and colleagues provide a useful framework for thinking about how these consequences manifest:

  - **Allocative harm**: when opportunities or resources are withheld from certain people or groups.

  - **Representational harm**: when certain people or groups are stigmatised or stereotyped.

- Algorithms that determine whether someone is offered a loan or a job risk inflicting **allocative harm**. This is the type of harm that we hear about, because it can be measured and is more commonly recognised as harmful.

- However, even if they do not directly withhold resources or opportunities, systems can still cause **representational harm**, such as language models that encode and replicate stereotypes.

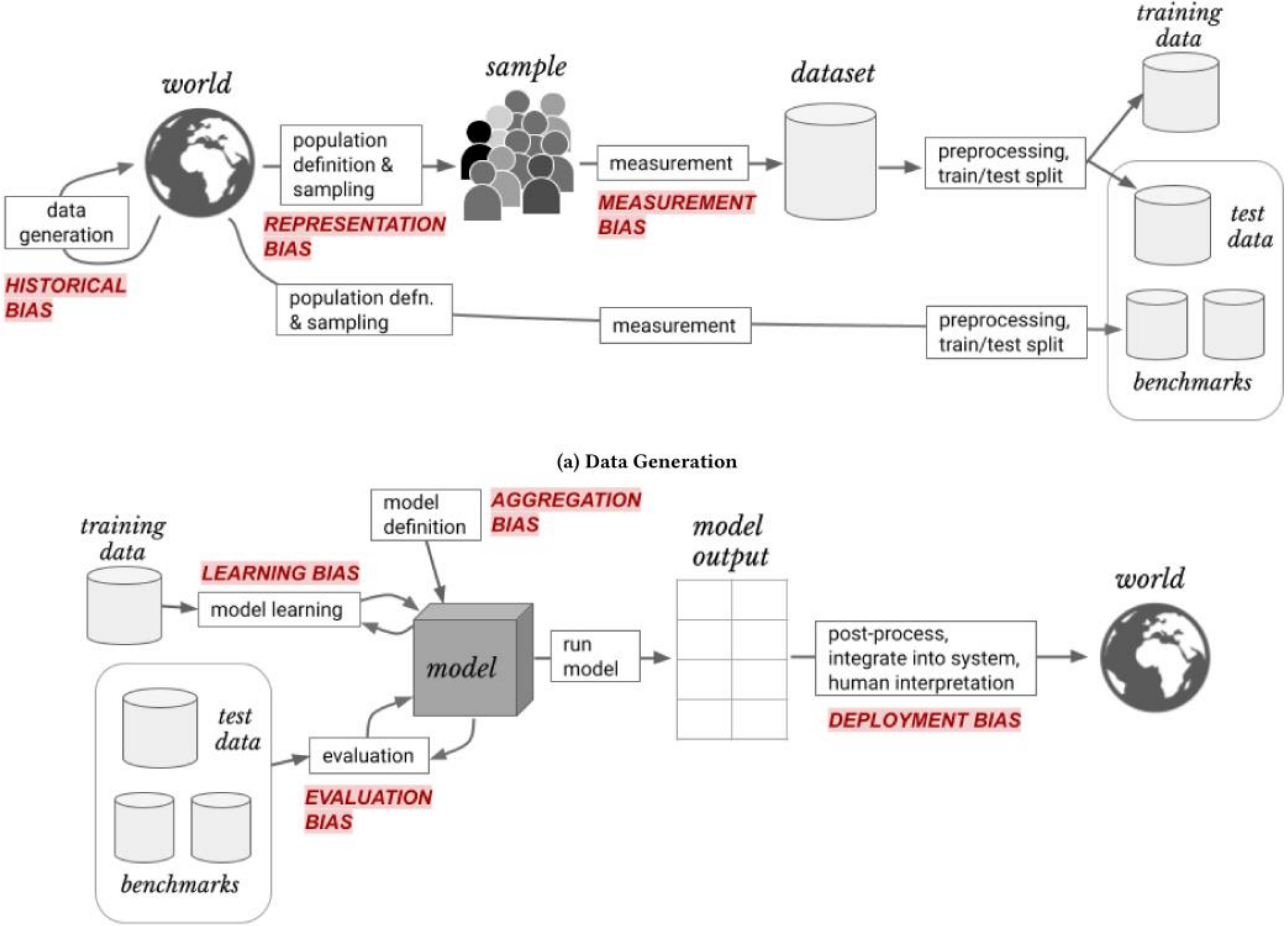
Suresh and Guttag (2021)

Awareness of ML's potential unwanted consequences

- To anticipate, prevent, and mitigate undesirable downstream consequences, it is critical that we understand when and how harm might be introduced throughout the ML life cycle.

- Suresh and Guttag (2021) provide a framework that identifies **seven distinct potential sources of downstream harm** in ML, spanning **data collection**, **development**, and **deployment**.

[Read Paper]

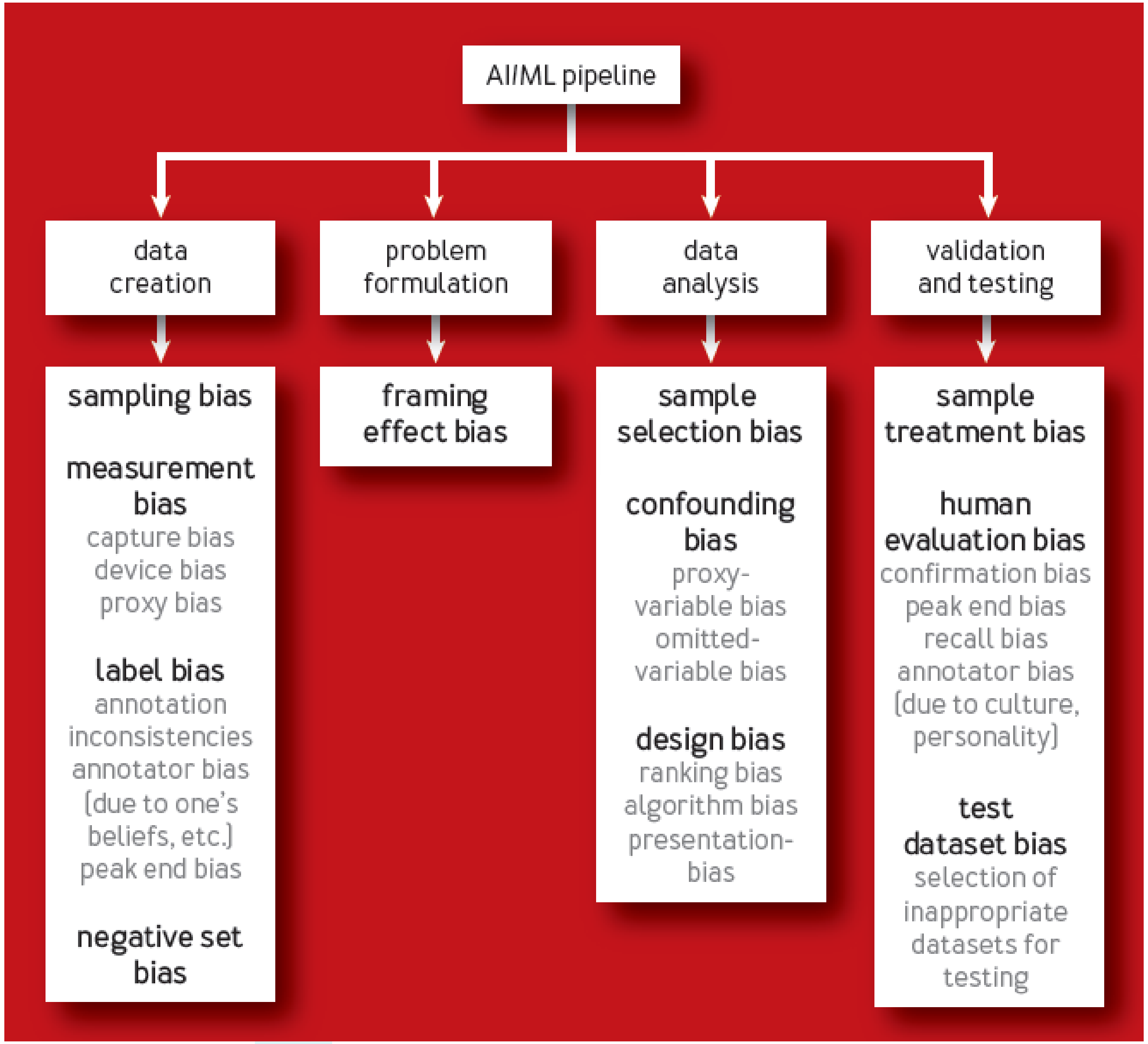


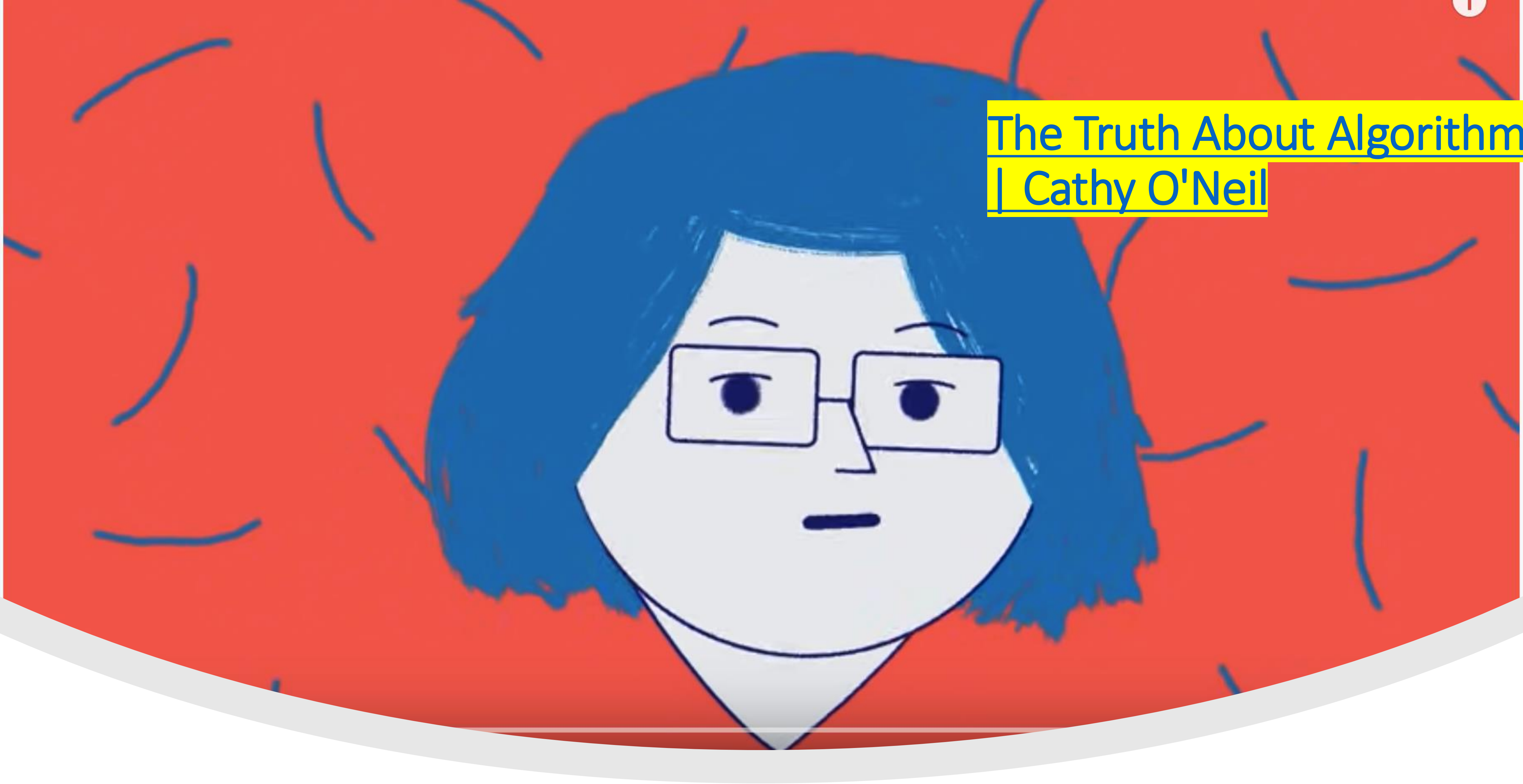
(a) Data Generation

(b) Model Building and Implementation

Biases in AI Systems

FIGURE 1: TAXONOMY OF BIAS TYPES ALONG THE AI PIPELINE

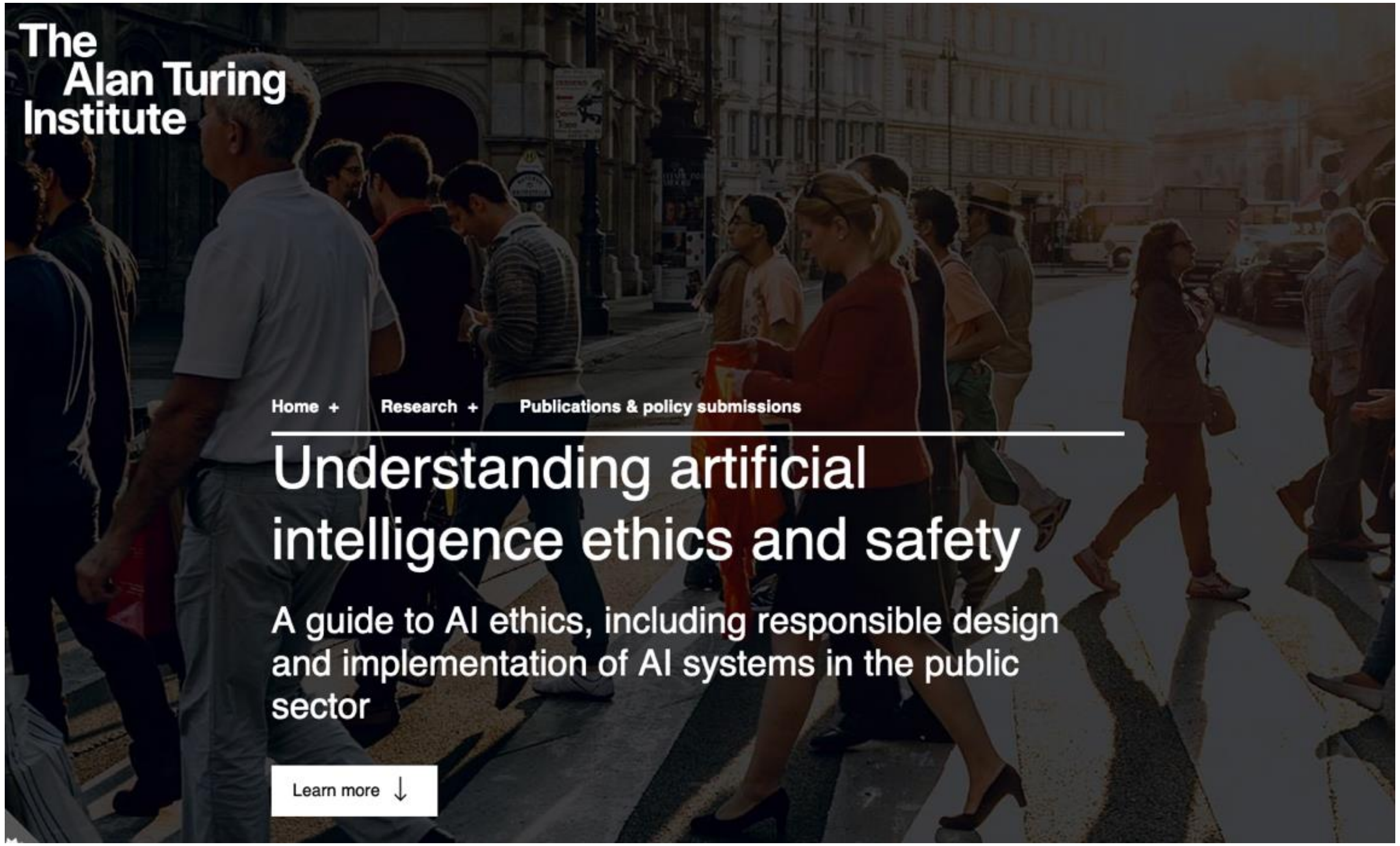




[The Truth About Algorithm | Cathy O'Neil]

- Some of AI’s applications are already questionable:
  - data collection that intrudes on privacy
  - facial recognition algorithms that are supposed to identify hostile behaviour or are with racial prejudice
  - military drones and autonomous lethal weapons
  - etc

- The ethical problems that AI raises – and will undoubtedly continue to raise tomorrow, with greater gravity – are numerous.


The Alan Turing Institute
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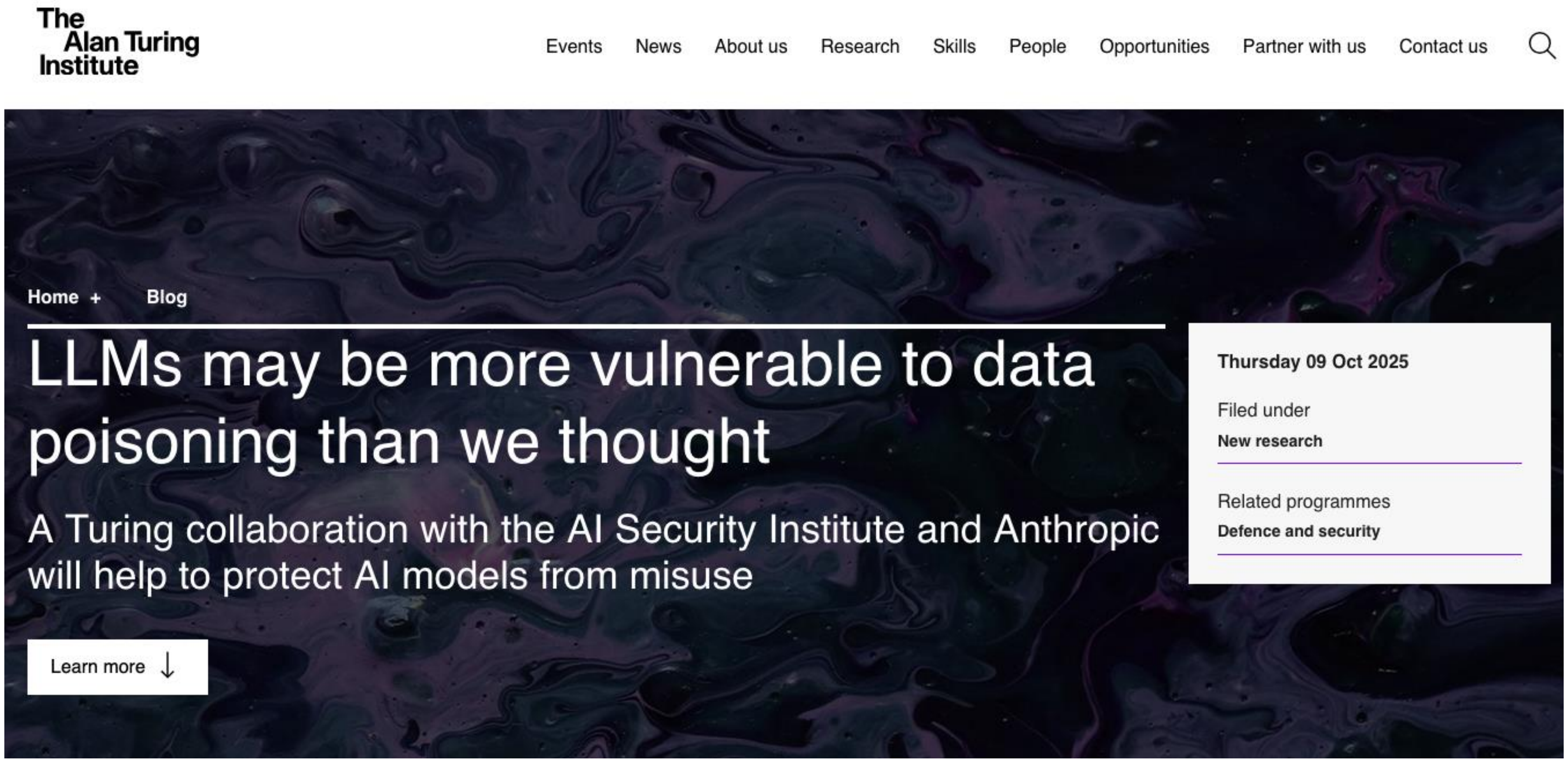
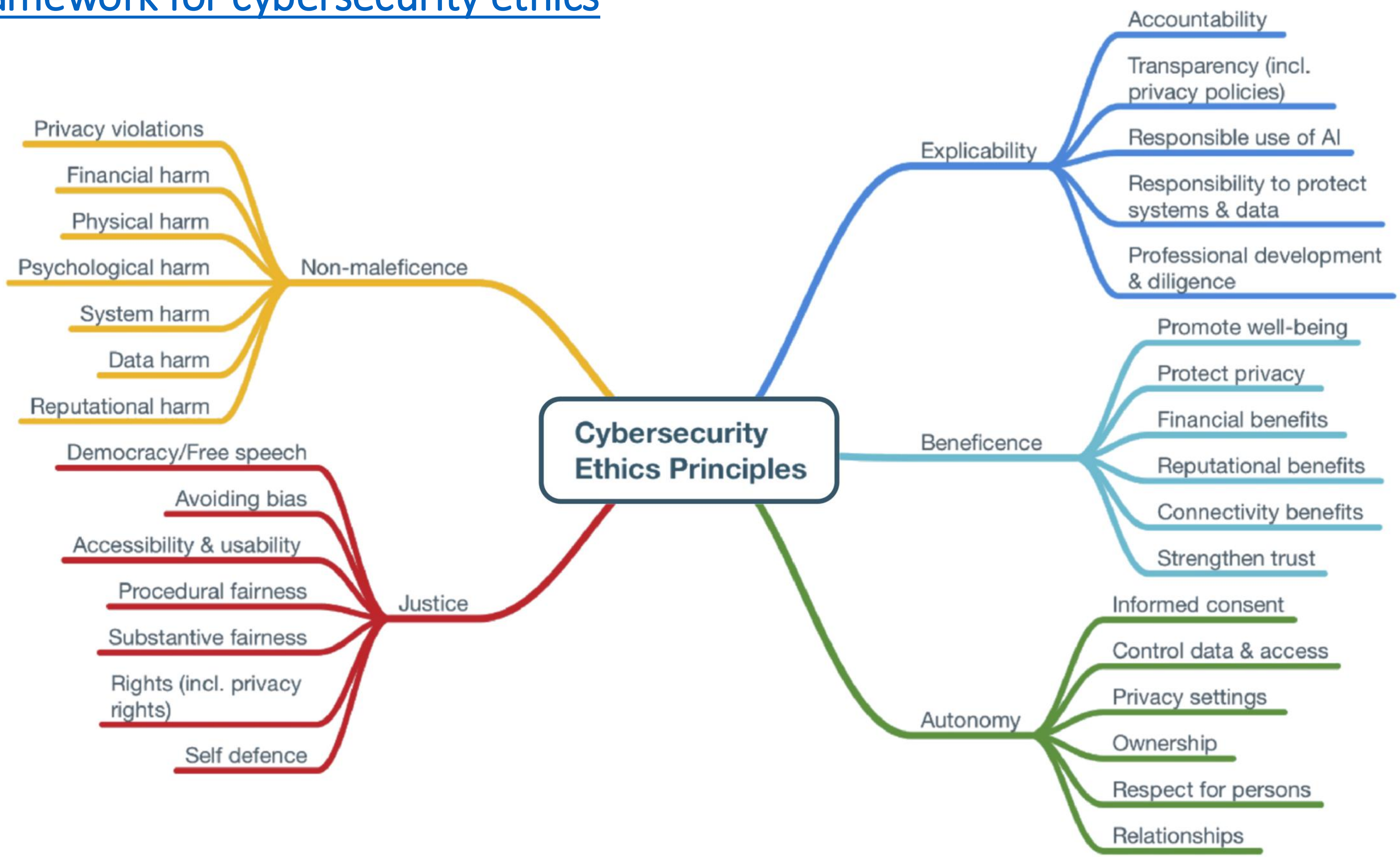
Research +
Publications & policy submissions
Understanding artificial intelligence ethics and safety
A guide to AI ethics, including responsible design and implementation of AI systems in the public sector
Learn more

The largest investigation of data poisoning yet conducted, finding that LLMs may be more vulnerable than previously thought.



The Alan Turing Institute

A principlist framework for cybersecurity ethics



See also Practical Cybersecurity Ethics

BCS Recommendations for IT Professionals

- Ensuring their professional development includes current ethical concerns.
- Strengthening specialist knowledge through appropriate accreditation and certification on ethical concerns.
- Contributing to the development of ethical standards and good practices.
- Supporting greater ethical and technical understanding of AI and other technologies among colleagues across their organisations and client-base, supported by their professional bodies.
- Using their subject expertise to engage with the broader public on issues of ethics of AI and digital technologies.

Ethics provides

- A framework for understanding and interpreting right and wrong in society.

- A set of standards for behaviour that helps us decide how we ought to act in a range of situations.

Research Ethics

- The moral principles that govern how researchers should carry out their work.
- These principles are used to shape research regulations agreed by groups such as university governing bodies, communities or governments.
- All researchers should follow any regulations that apply to their work.

University Code of Practice

“

Research projects being undertaken by taught UG and PG students as part of their degree should be discussed in detail between the student and the supervisor. For these research projects the supervisor will act as the Principal Investigator and will be responsible for ensuring ethical standards are met and for ensuring ethical approval and/or management approval(s) are sought by the student researcher where appropriate.

[Code of Practice Governing the Ethical Conduct of Research 2020-21]

Session Part II

Legal Issues

UK Legislation

- [Data Use and Access Act 2025]

    The Data (Use and Access) Act 2025 reforms how the UK manages non-personal and personal data and aims to unlock the secure and effective use of data. This wide-ranging Act enables:

    - the growth of digital verification services
    - new Smart Data schemes like Open Banking
    - a new National Underground Asset Register
    - important changes to the UK's data protection and privacy legislation

- The Data Protection Act 2018
- The Computer Misuse Act 1990
- The Copyright, Designs and Patents Act 1988
- Creative Commons Licensing
- The Freedom of Information Act 2000

Is there UK legislation on AI?

In February 2024, the government announced a range of measures in support of its approach—building upon the white paper it issued on the regulation of AI in 2023—including the first iteration of guidance to regulators which includes voluntary regulatory principles.

[Artificial Intelligence (Regulation) Bill [HL] - UK Parliament]()

IEEE/ISO/IEC 24748-7000-2022

IEEE/ISO/IEC International Standard--Systems and software engineering--Life cycle management--Part 7000: Standard model process for addressing ethical concerns during system design

Purchase | Access via Subscription

Active Standard

A set of processes by which organizations can include consideration of ethical values throughout the stages of concept exploration and development is established by this standard. Management and engineering in transparent communication with selected stakeholders for ethical values elicitation and prioritization is supported by this standard, involving traceability of ethical values through an operational concept, value propositions, and value dispositions in the system design. Processes that provide for traceability of ethical values in the concept of operations, ethical requirements, and ethical risk-based design are described in the standard. All sizes and types of organizations using their own life cycle models are relevant to this standard.

Session Part III

What you need to do for the UG Ethics Approval Process

How does this apply to me?

- When you are planning your final project you should spend some time thinking about its ethical implications:
    - thinking about what you might need to do to protect people who take part and/or the environment
- Follow the UG Ethics Approval Process

UG Project Students

- All UG students undertaking their final project must consider the ethical aspects of their work in line with University procedures.
- **Please read carefully** through the University's Ethics Code of Practice

To summarise: there are 2 Forms

- **Part A:** Students should begin the process of consideration of research ethics approval by completing the Part A application form.

- If there are minimal ethical implications – see the decision tool at the end of Part A - then they will not need to complete the Part B application form.

- **Part B:** If the research raises ethical issues, the student will be asked to complete Part B of the application.

Where can I find the forms?

- You are only able to find Part A form
- See this module's BB for the link to access the form.
- Please check with your supervisor if your project needs Part B

What software tools are students allowed to use for data collection?

- MS Forms

You also have access to MS Forms through your University of Westminster account.

You may wish to use video and audio-conferencing software to interview participants for your research. The University’s preferred real-time video and audio conferencing platform is Microsoft Teams.

It is advised that you do NOT record such sessions.

According to the School

> *“we do not restrict students if they wish to use different platforms. Some students conduct interviews face-to-face or on-line with potential customers, etc.”*

Where should I save my data?

- OneDrive

The fundamental ethical principle is that there should be no risk of harm through participating in research, and particular care needs to be taken with participants such as children and vulnerable adults.

Also, a decision to participate should be based on informed consent.

Full information must be provided in a Participant Information Sheet (PIS), and, after reading this, a participant should normally sign a Consent Form.

- Begin the process of consideration of research ethics approval by completing the *Part A application form*. If there are minimal ethical implications – see the decision tool at the end of *Part A* - then you will not need to complete the Part B application form.
- In this case, your application does not need to be submitted to the College Research Ethics Committee.
- Once you complete the form contact your supervisor for approval.

- You need to consider carefully your methodology, any areas of participant questioning (if applicable), potential risks, etc. This should be done together with your supervisor. <u>Please note that your supervisor is there to guide and advise you and not in order to do the work for you.</u>

- You need to ensure that you have adequately thought through the research methodology, any areas of participant questioning (if applicable), potential risks, etc. before submitting an application. If full ethics approval is required, the documentation will be submitted to the VRE by your supervisor.

- *Part B application form* will need to be completed if your study has ethical implications. Then the *Parts A and B, PIS*, and *Consent Form*, should be put into one file, create a zip file and upload it on Blackboard.
- Email your supervisor for approval.
- Your Supervisor will then submit your application on VRE for consideration to the appropriate ethics Committee.
- You should bear in mind that there is likely to be some delay before a decision is made by a Committee.

Valid Consent and Participant Information

- Valid and appropriate consent should be obtained orally or in writing and must be documented before any research can begin. If oral consent is being sought, the Principal Investigator must ensure it is documented and a reason for not gaining written consent must be provided to a Research Ethics Committee as appropriate.
- For research involving techniques such as internet surveys, journalistic interviews or market research, for example, other approaches to documenting consent may be used, in consultation with relevant professional codes e.g. Code of Conduct of the Market Research Society.

Code of Practice Governing the Ethical Conduct of Research 2020-2021 (section 10)

Valid Consent

- Valid and appropriate consent in this Code of Practice is defined as: participant's consent (or the Representative of a deceased donor) given freely and independently, in the absence of coercion, in light of information provided to the participant.
- Principal Investigators are required to inform participants or the Representative of a deceased donor about anything that could affect their decision to take part. Consent may need to be an ongoing process for participants of a study

Code of Practice Governing the Ethical Conduct of Research 2020-2021 (section 10)

Some participants may lack the ability to give their informed consent to participate in research, for example:

- **Children:** If school children are asked to be participants in a school-based environment, the Principal Investigator shall inform the Head Teacher of the school of what is proposed and obtain their permission for pupils to take part. The parent(s) or guardian(s) consent should also be sought. Such permission is in addition to, not instead of, individual consent previously described. Minors must be informed that they have the same rights as an adult.

Participant Information Sheet (PIS)

it should inform the participant of the following in plain, jargon-free language:

- why they have been chosen as a potential participant
- the aims of the research and why it is being undertaken
- whether the research is part of a student project and/or the University of Westminster affiliation
- exactly what the participant is required to do
- whether there is an inclusion or exclusion criteria and what this is any harm which might occur as a result of participation
- the right to complain, and to whom, in the event of a problem or perceived issue in the research study or with the research team
- the right to withdraw, or withdraw their data, from the investigation as practicable
- arrangements ensuring the confidentiality and privacy of the participant and protection of the data
- technical protection of the data
- what will happen to their data after the research, e.g. destruction, archiving, etc. and the relevant timescales involved taking account of any requirements to retain data for formal audit purposes
- contact details of the Supervisor for further questions and to report adverse or serious events
- the requirement to report any symptoms which may occur
- how the participant will be informed of the results of the research if applicable
- the intended use(s) of the results of the research
- how the research will be published or disseminated.
- any limitations of confidentiality
- consent for future research

"*A copy of the Participant Information Sheet must be retained together with the signed Consent Form and stored suitably in the records of the investigation. A copy of the Participation Information Sheet should also be made available for the participant to take away.*

Forms and resources

- See BB for links and information.

- <u>For any sensitive data (e.g. medical, sexual life, religious or political views, etc.) online tools should probably not be an option for a UG project</u>

You must check with your supervisor BEFORE you plan any online data collection via surveys or interviews.

You must check with your supervisor first and follow the University policy BEFORE proceeding with any recordings.

Data Security and Confidentiality

- Relevant Data Protection legislation and University guidance in data security must be observed in the collection, use, storage, back-up and eventual destruction of all data.

Ensuring Anonymity (examples)

- The respondents are not required to give their identification like name or address.
- Further ways of achieving anonymity could be a) the use of aliases or b) the use of codes for identifying people (to keep the information on individuals separate from access to them).
- It is important to include in the questionnaire assurances of confidentiality, anonymity and non-traceability. This, for example, could be done by indicating that respondents do not need to give their name, that the data will be aggregated, that individuals will not be able to be identified through the use of categories or details of their location etc.

Guidelines on security arrangements for any recorded information (1/3)

1. Any recorded information should be adhered to the Data Protection Act 2018 requires that the participant's informed consent be sought where personal information is to be used and that those who have access to the information, or receive copies of it, are clearly identified.
2. Systems used for the storage of data should ideally be located on University Information Services' secure network infrastructure within the firewall so that access control measures and auditing policies can been forced.
3. Desktop/laptops used by researchers should always be fully patched and ideally, regularly scanned for software vulnerabilities.

Guidelines on security arrangements for any recorded information (2/3)

4. All the data held on recordable media (e.g. discs, USB storage devices) should be password protected as a minimum security measure, to protect the contents if they are lost.
5. Any recordable media containing identifiable personal data should be stored securely when not in use.
6. Knowledge of procedures and passwords to access any medical or research data of named individuals should be held securely and be made available only to those authorised.

Guidelines on security arrangements for any recorded information (3/3)

7. File protection (Encryption) should be in operation on computer systems used to hold named individual data.
8. Transmission of identifiable personal data across public communication lines (e.g. Email, DropBox etc.) should be avoided at all times. Where this is absolutely necessary, the prior approval of a Research Ethics Committee is required.
9. Access to the data should be directly supervised by a designated system manager and permitted only to those authorised.

Data Secure Disposal/Destruction

- Once the project is complete and following the examination (and results), consideration must be given to the secure disposal and destruction of any data that is either, provided at the outset or gathered during the project.
- Where personal information is involved, then specific measures to ensure the data is securely 'deleted' must be implemented.

Risk Assessment

- Please discuss with your supervisor the possibility that a Risk Assessment Form needs to be completed for your project.

Guide to the General Data Protection Regulation (GDPR)

The GDPR forms part of the data protection regime in the UK, together with the new Data Protection Act 2018 (DPA 2018).

Data Protection Act (2018)

EU General Data Protection Regulation (GDPR)

- http://www.eugdpr.org
- It is important to keep in mind that GDPR requires both a *‘controller’* and a *‘processor’* of personal data to consider due diligence when processing personal data, and both are now equally responsible for lawful processing.
- https://ec.europa.eu/commission/priorities/justice-and-fundamental-rights/data-protection/2018-reform-eu-data-protection-rules_en

What is Personal Data?

Personal data is defined in the GDPR as:

- *"'**personal data'** means any information relating to an identified or identifiable natural person ('data subject'); an identifiable natural person is one who can be identified, directly or indirectly, in particular by reference to an identifier such as a name, an identification number, location data, an online identifier or to one or more factors specific to the physical, physiological, genetic, mental, economic, cultural or social identity of that natural person".*
- https://ico.org.uk/for-organisations/guide-to-the-general-data-protection-regulation-gdpr/what-is-personal-data/what-is-personal-data

Further Resources

- Ethical concerns about facial recognition systems
- **ACM Code of Ethics and Professional Conduct**
- **IEEE Code of Ethics**
- **BCS Code of Conduct**
- **Ethics Guidelines for Internet-Mediated Research (2021))**

- **Information Commissioner's Office**
- **https://ico.org.uk**
- The UK's independent authority set up to uphold information rights in the public interest, promoting openness by public bodies and data privacy for individuals.

Recommended read:
BCS Report

[Living with AI and emerging technologies: Meeting ethical challenges through professional standards]

Consider..

- Where an individual intentionally keeps any information about their identity private, at least one ethical question arises: **what right do others have to make it public?**
- [Davis, K. (2012). *Ethics of Big Data: Balancing risk and innovation*. O'Reilly Media, Inc.]

"Research Ethics is an ongoing consideration and needs to be considered, understood and applied by the researcher to the entire research lifecycle, revisited as appropriate and intermittently, including from inception, proposal, data collection, writing, publication and dissemination of results.

Thank you & Good Luck!

Frantzeska Kolyda MPhil SFHEA FBCS